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proc import
datafile="D:\employee.csv"
out=aa1
dbms=csv;
run;

/*資料位置性質概況 */
proc contents data=aa1;
run;

/*資料內容列印5筆 */
proc print data=aa1(obs=5);
run;

/*次數分配表 */
proc freq data=aa1;
    tables gender jobcat overtime;
run;

/*畫直方圖 */
proc chart data=aa1;
    vbar salary/midpoints=20000 to 50000 by 5000;
run;

/*計算平均數 */
proc means data=aa1;
    var salary salbegin jobtime prevexp;
run;

/*與敘述統計相關之所有分析 */
proc univariate data=aa1 normal plot;
    var salary;
run;

/* one sample t test */
proc ttest h0=30000 alpha=0.05;
    var salary;
run;

/* two sample t test */
proc ttest alpha=0.05;
    class gender;
    var salary;
run;
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/* Mann-Whitney U test */
proc npar1way data=aal;
class gender;
var salary;
run;

/*交叉分析表 */
proc freq;
tables gender jobcat;
tables gender*jobcat /chisq norow nocol noperc;
run;

/* ANOVA */
proc anova data=aal;
class gender jobcat;
model salary = gender jobcat;
run;

/* Kruskal-Wallis K test */
proc npar1way data=aal;
var salary;
class jobcat;
run;

/* scatter plot */
proc plot;
plot salary * salbegin;
run;

/* correlation coefficient matrix */
proc corr;
var salary salbegin jobtime prevexp;

/* regression analysis */
proc reg data= aal;
model salary = salbegin /p cli r;
plot r.*salbegin;
plot r.*p.;
run;

/* logistic regression analysis */
proc logistic data=aal order=data;
class jobcat;
model overtime = jobcat jobtime;
run;

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/*mixed model*/
data machine;
input machine person rating @@;
datalines;
1 1 52.0 1 2 51.8 1 2 52.8 1 3 60.0 1 4 51.1 1 4 52.3 1 5 50.9
1 5 51.8 1 5 51.4 1 6 46.4 1 6 44.8 1 6 49.2 2 1 64.0 2 2 59.7
2 2 60.0 2 2 59.0 2 3 68.6 2 3 65.8 2 4 63.2 2 4 62.8 2 4 62.2
2 5 64.8 2 5 65.0 2 6 43.7 2 6 44.2 2 6 43.0 3 1 67.5 3 1 67.2
3 1 66.9 3 2 61.5 3 2 61.7 3 2 62.3 3 3 70.8 3 3 70.6 3 3 71.0
3 4 64.1 3 4 66.2 3 4 64.0 3 5 72.1 3 5 72.0 3 5 71.1 3 6 62.0
3 6 61.4 3 6 60.5
;
proc glm data=machine;
class machine person;
model rating=machine person machine*person;
random person machine*person / test;
run;

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data factor;
input id a b c d;
datalines;
1 11 9 7 4
2 9 8 11 13
3 9 11 5 4
4 8 13 15 14
5 6 5 6 11
6 2 6 9 9
7 5 7 4 8
8 12 11 4 5
9 15 12 11 13
10 1 6 4 6
;
/* principle component analysis */
proc princomp;
var a b c d;
run;

/* factor analysis */
proc factor method=p priors=smc msa n=2
simple corr res;
var a b c d;
run ;

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/* Cox regression*/
data renal;
input group id renal days censor @@;
datalines;
1 1 1 8 1 2 13 1 13 1 1 12 0 52 1 2 16 1 18 1 1 5 1 58 1 2 25 1 23 1 1 8 1 63 1
2 11 0 70 1 1 21 1 63 1 2 10 0 76 1 1 7 0 220 1 2 2 0 180 1 1 24 0 365 1
2 9 0 195 1 1 4 0 452 1 2 20 0 210 1 1 18 0 496 1 2 3 0 232 1 1 22 0 528 0
2 17 0 300 1 1 19 0 560 0 2 23 0 396 1 1 15 0 676 0 2 14 0 490 0 2 6 0 540 0
;
proc phreg data=renal;
model days*censor(0)=group renal;
run;

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/* Generalized Estimating Equation*/
proc import
datafile="D:\anxiety.csv"
out=cc1
dbms=csv;
run;

proc genmod;
class id;
model score=anxiety tension trial/dist=norm link=identity type3;
repeated subject=id/type=ar(1)corr;
run;

```